

KADEN N. NGUYEN

(703)-649-1929 | kadennguyen0329@gmail.com

[linkedin.com/in/kadennguyen0329](https://www.linkedin.com/in/kadennguyen0329) | github.com/kadennguyen0329

EDUCATION

University of Virginia

B.A. in Computer Science; Minor in Data Science

GPA: 3.87/4.0; Dean's List 2023-2025

Relevant Coursework: Software Engineering, Software Testing, Cloud Computing, Artificial Intelligence, Data Science Systems

Charlottesville, VA

August 2023 – May 2026

SKILLS

Languages: Java, Python, C, C#, SQL

Frameworks & Libraries: Django, React.js, Spring Boot, Scikit-learn, NumPy, Pandas, JUnit, OpenSearch

Tools: Linux, PostgreSQL, MySQL, SQLite, MongoDB, Git, Docker, AWS S3, EC2, Heroku, Gradle, Maven, Jira

Concepts: Agile (Scrum), Distributed systems, Test driven development, Performance testing, CI/CD, RESTful APIs

WORK EXPERIENCE

Comcast (FreeWheel)

Reston, VA

Software Engineer Intern

June 2025 – August 2025

- Worked on production-grade distributed service for querying advertising data designed to handle over 12 million queries a day with >99.99% uptime, participating in an agile team with two-week Scrum sprints, sprint planning, and daily standups
- Designed and implemented semantic search functionality for advertising data using OpenSearch's Java client, enabling conceptually relevant results through vector-based matching and improving user query accuracy by ~20%.
- Conducted performance benchmarking across multiple text-embedding models, performing big data ingestion (>1 million documents) and executing 10,000+ search queries to optimize semantic search speed and reduce query times by ~50%.
- Orchestrated end-to-end testing and documentation efforts, writing comprehensive unit and integration tests (90% code coverage) and internal wiki pages to ensure robustness, edge case coverage, and team knowledge sharing.

UVA Department of Computer Science

Charlottesville, VA

Teaching Assistant

January 2025 – May 2025

- Facilitated instruction in Computer Systems and Organization I, covering core topics such as circuits, computer architecture, instruction set architecture (ISA), assembly language, and C programming.
- Directed in-person lab sessions and held office hours for 30+ students, providing individualized support and clarifying complex concepts to ensure understanding and successful assignment completion.

Techsur Solutions

Herndon, VA

Software Engineer Intern

June 2024 – August 2024

- Designed and implemented a dynamic React.js user interface to enable advanced filtering and sorting of 7,000+ uploaded documents, resulting in a ~60% improvement in document processing speed and enhanced operational efficiency.
- Styled and optimized the front-end of a chatbot application using SCSS, aligning with visual design specifications and delivering a ~25% increase in user satisfaction, as measured by feedback from 10+ usability testers.
- Developed a RESTful API using Spring Boot for a task management microservice for streamlining task workflows.

UVA Omni-Reality & Cognition Lab

Charlottesville, VA

Research Assistant

January 2024 – May 2024

- Collaborated with Professor Arsalan Heydarian on Department of Transportation-funded research, using Unity3D to construct immersive VR simulations of Charlottesville cycling environments, accelerating behavioral data collection by ~80%.
- Applied Unity ProBuilder and C# scripting to dynamically model and manipulate simulation elements (e.g., bike lanes, roundabouts), improving research efficiency by ~30% through controlled experimental setups.

PROJECTS

Airdrobe

- Led a 6-person team as Scrum Master, managing sprint planning, task delegation, and daily standups to deliver a peer-to-peer clothing lending platform with advanced features including multi-user support, personalized dashboards, and dynamic filters.
- Programmed application in Django, enforced data persistence with PostgreSQL and AWS S3, integrated Google OAuth, and deployed via Heroku with CI/CD pipelines in GitHub Actions.

Drug Use Prediction with Machine Learning

- Performed exploratory data analysis and PCA to identify behavioral clusters; cleaned and engineered features using Pandas and NumPy for model-ready inputs.
- Tuned and evaluated multiple scikit-learn models to predict cannabis usage based on personality traits, achieving a 61.97% F1 score with K-Nearest Neighbors.